# 10397
# Connect the Campus

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# 題目大意

學校要在各大樓間增設電纜，在不能改動線有電纜的條件下，希望能以最少費用安裝完成。

| INPUT | | /OUTPUT |
|---|---|---|
| 4 | 大樓數量 | 4.41 |
| 103 104 | 大樓座標 | 還需要的電纜長 |
| 104 100 | | |
| 104 103 | | |
| 100 100 | | |
| 1 | 已有電纜數量 | |
| 4 2 | 已連接的大樓 | |

＊電纜之間沒有方向性

＊建築物編號從１開始

＊每對建築物只需要一根電纜連接

＊沒有兩個建築物在同一個座標點

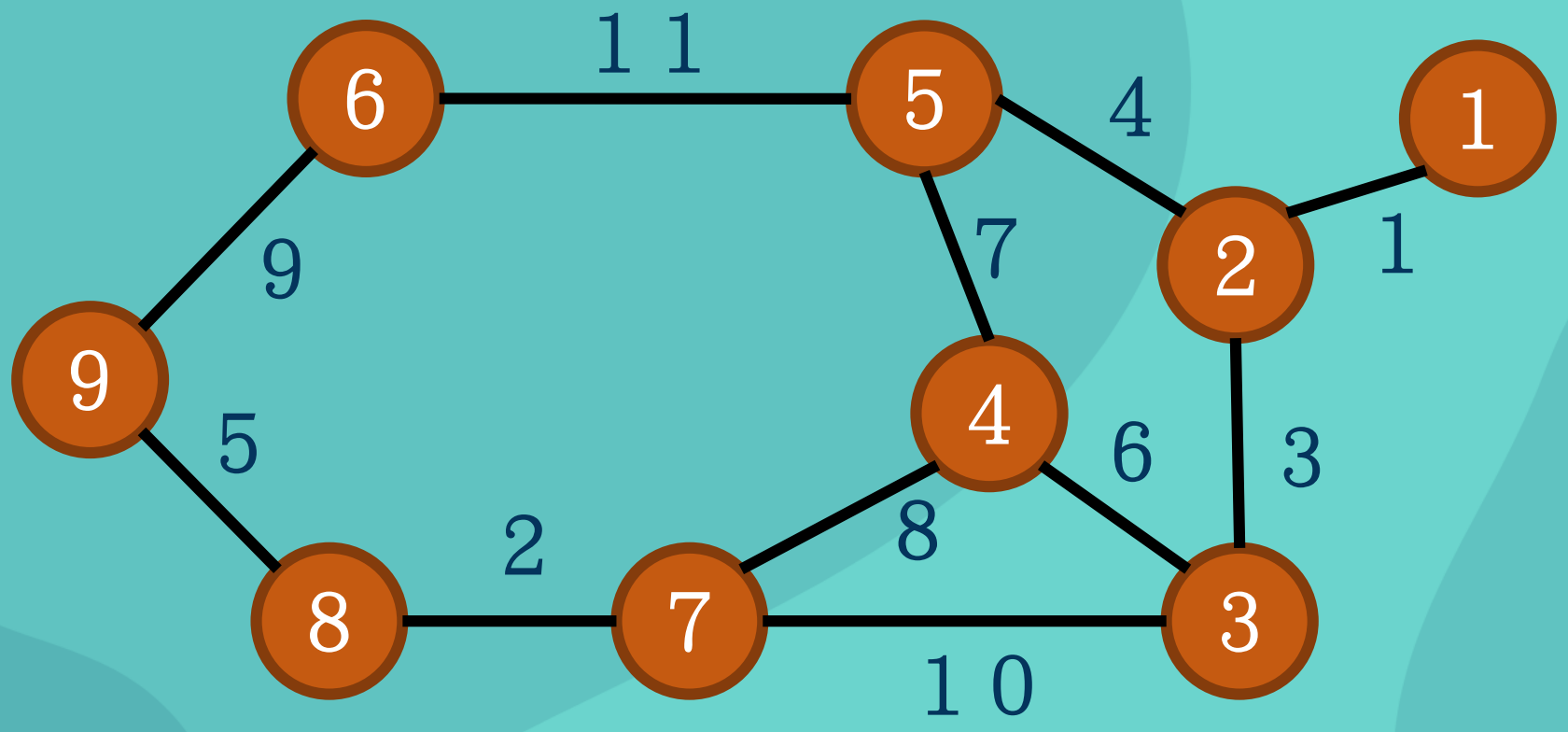
# Minimum Spanning Tree (greedy)

1. 新建圖，圖中擁有原圖中相同的節點，但沒有邊
2. 在原圖中加入所有可能邊，使其成為complete graph，並將原圖中所有的邊按權重從小到大排序
3. 從權重最小的邊開始，如果這條邊連接的兩個節點於圖G中不在同一個連通塊中，則添加這條邊到圖$\mathscr{G}$中
4. 重複3，直至圖$\mathscr{G}$中所有的節點都在同一個連通分量中

*連通塊：擁有不同的 Ancestor

# Minimum Spanning Tree (greedy)

＊白數字為建築物編號
＊黑數字為邊長排序
（由短至長）



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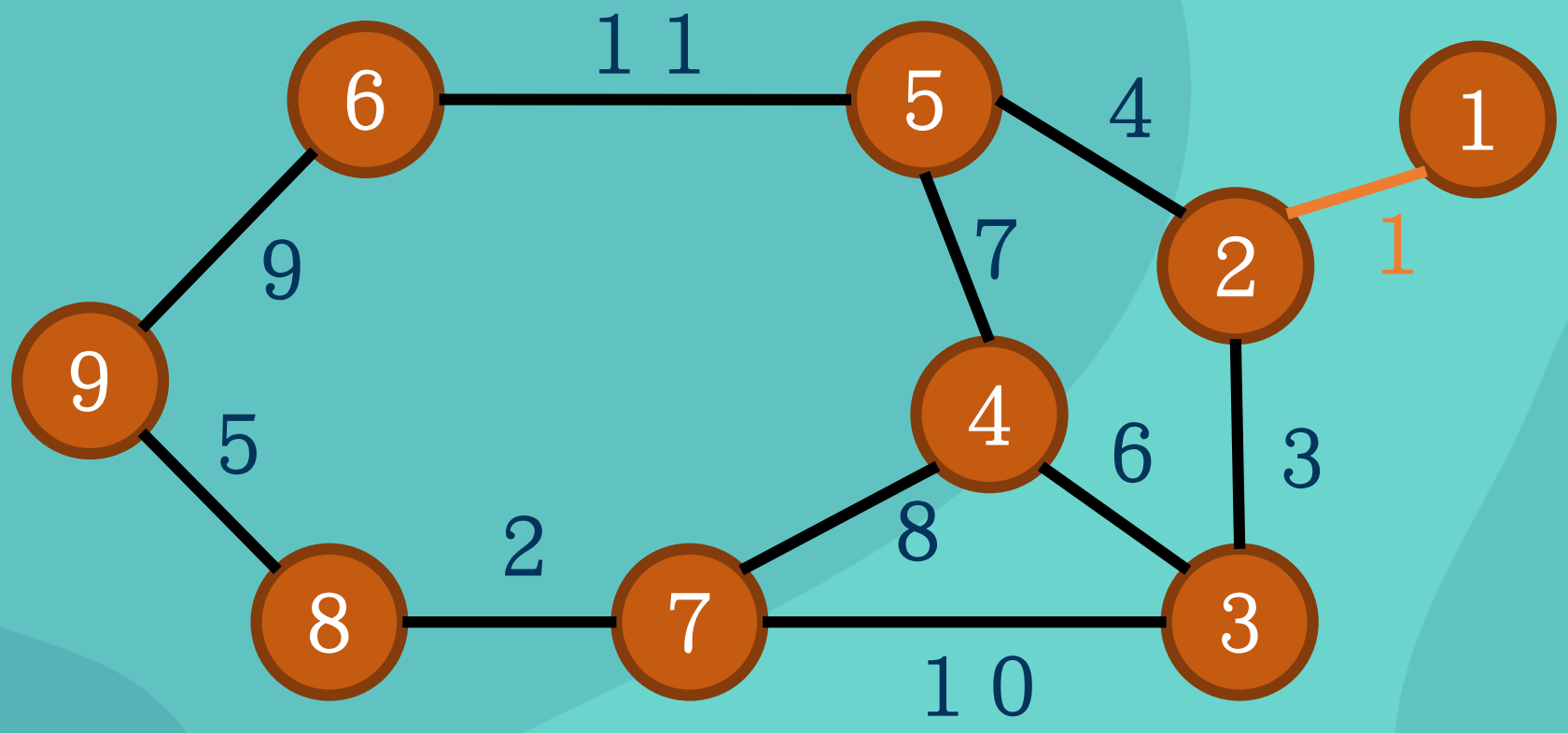

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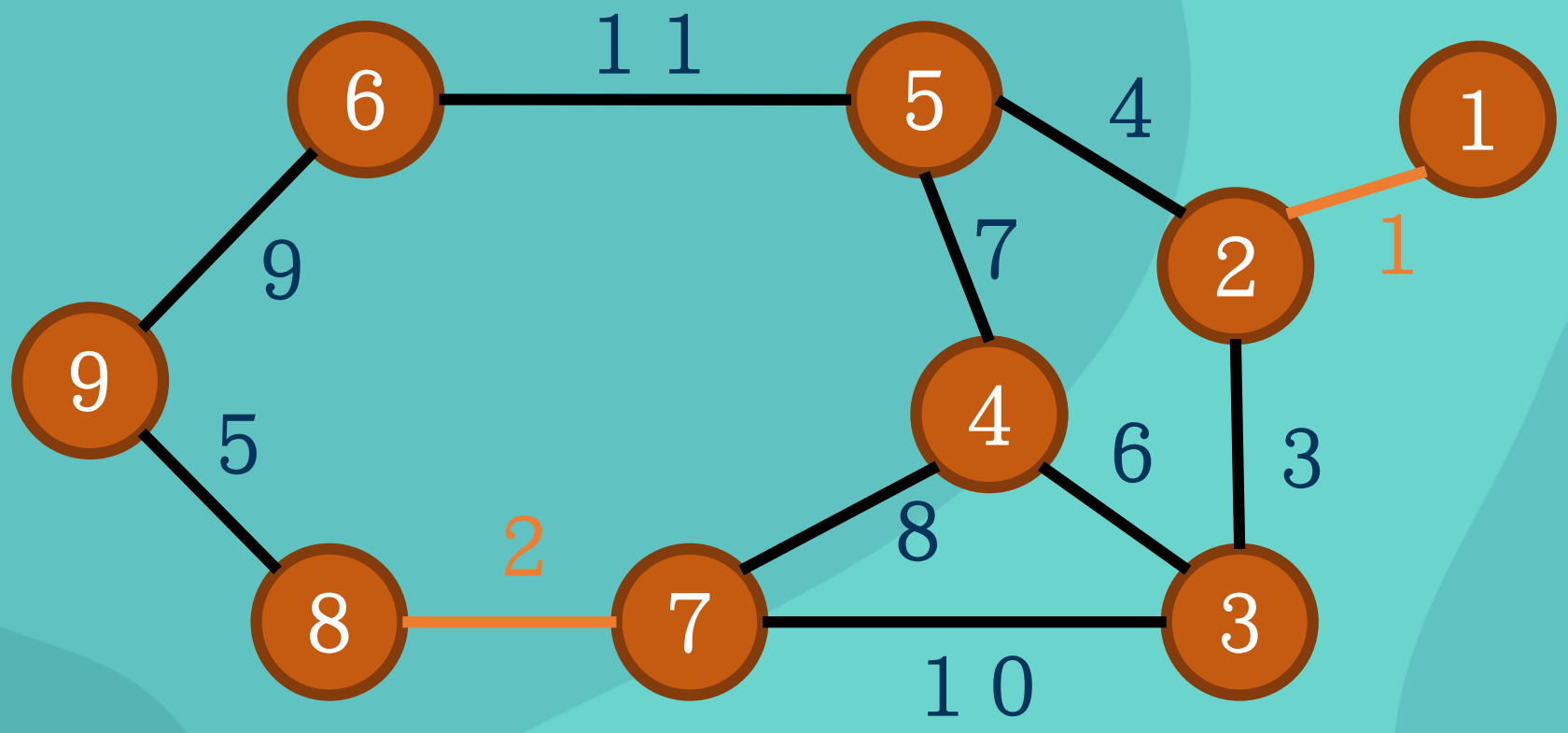

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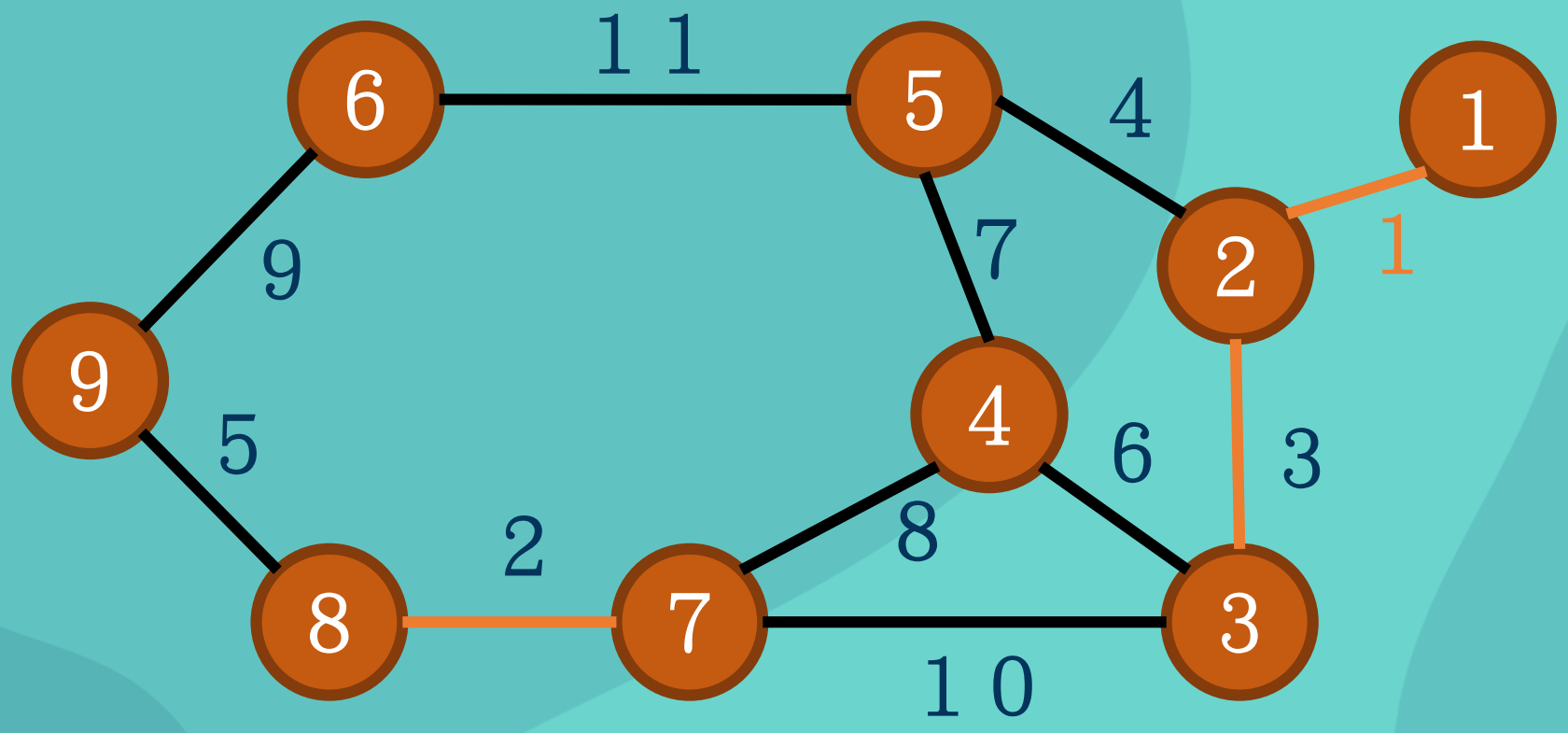

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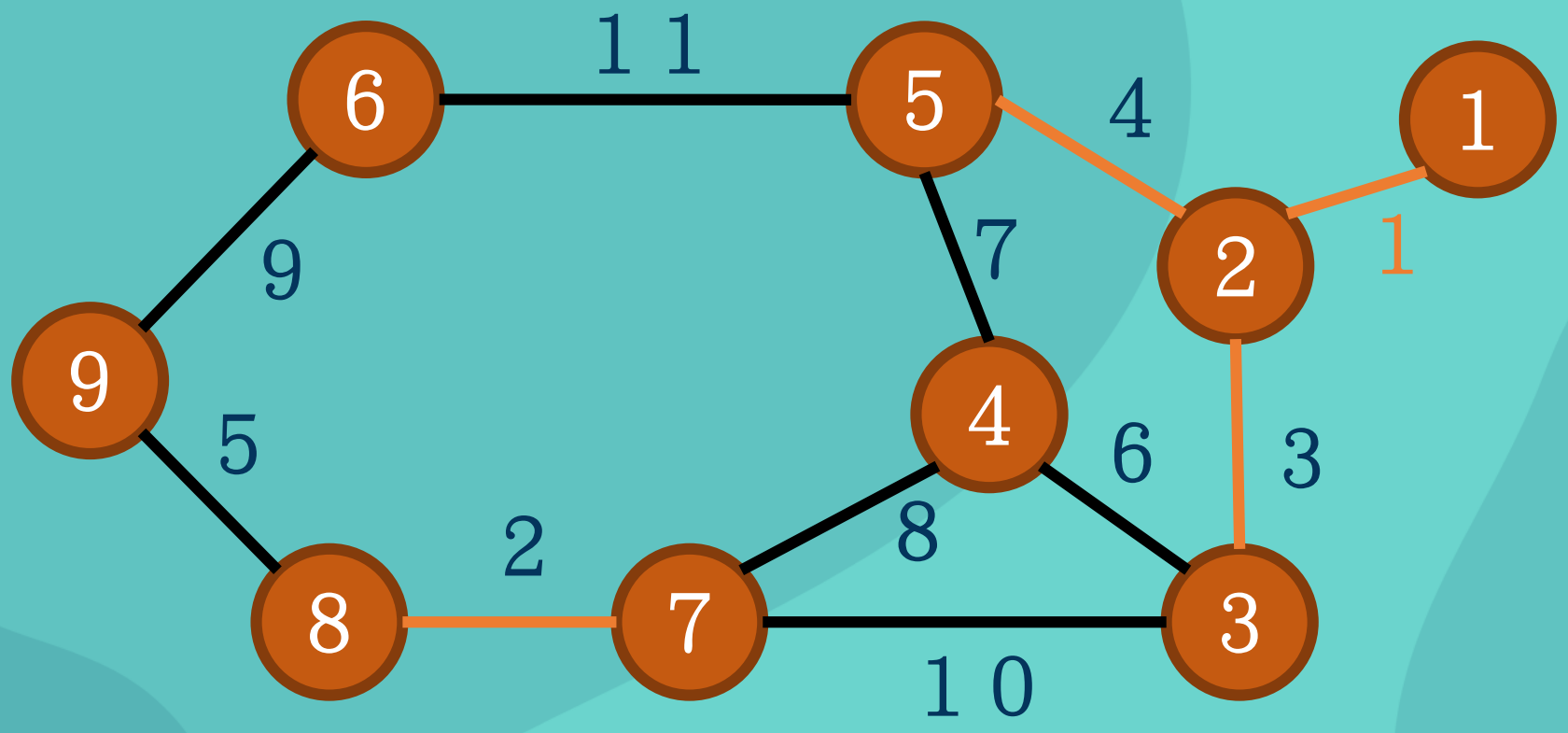

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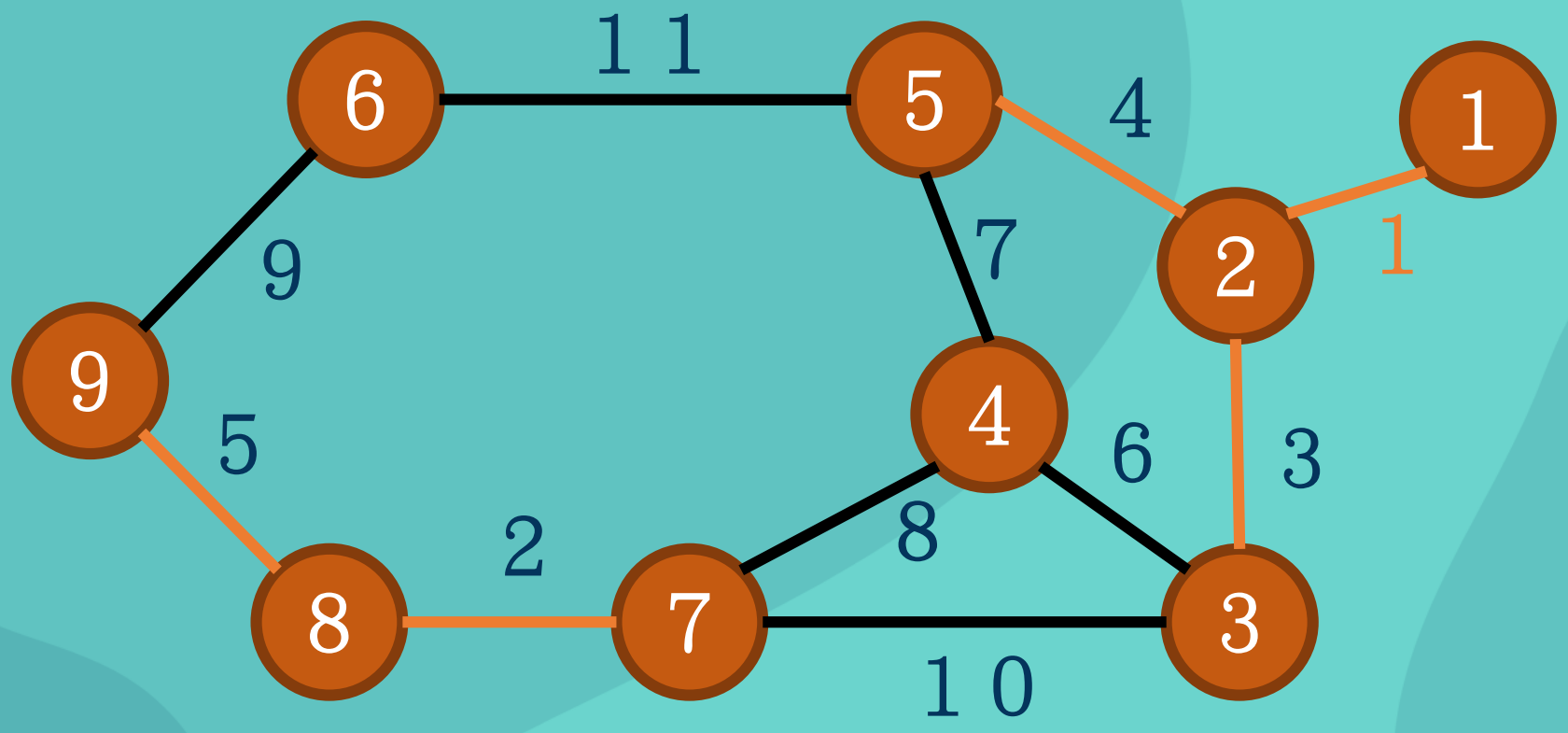

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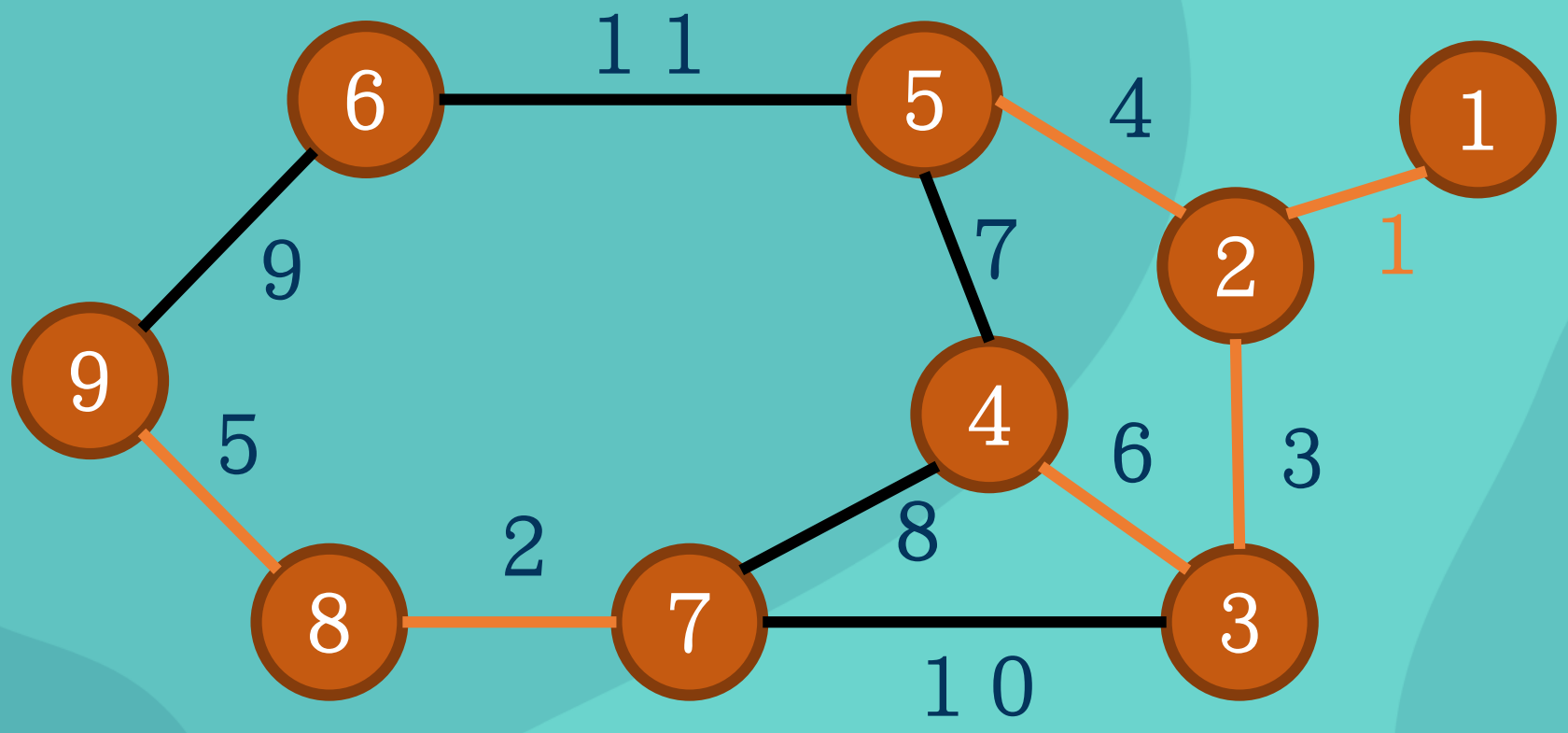

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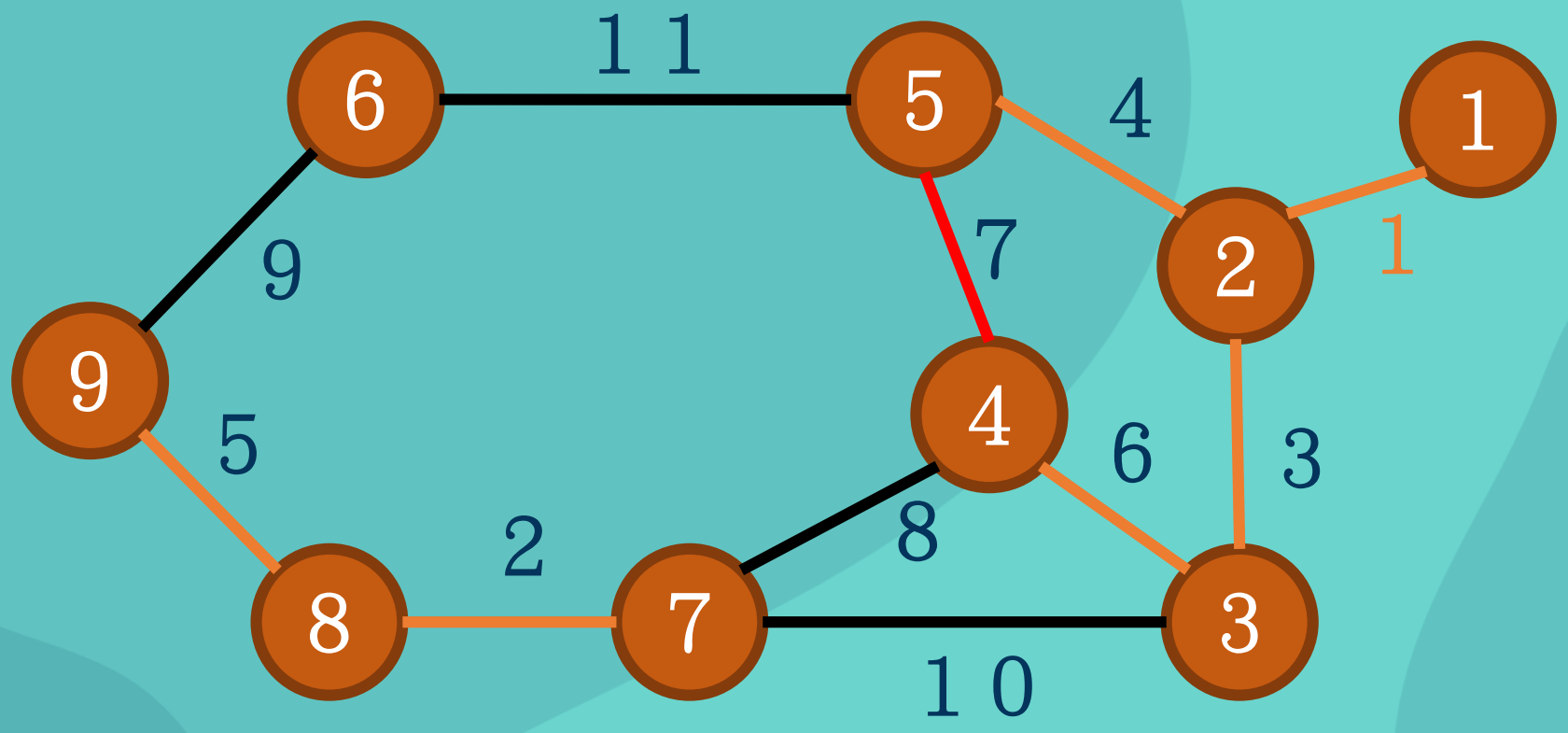

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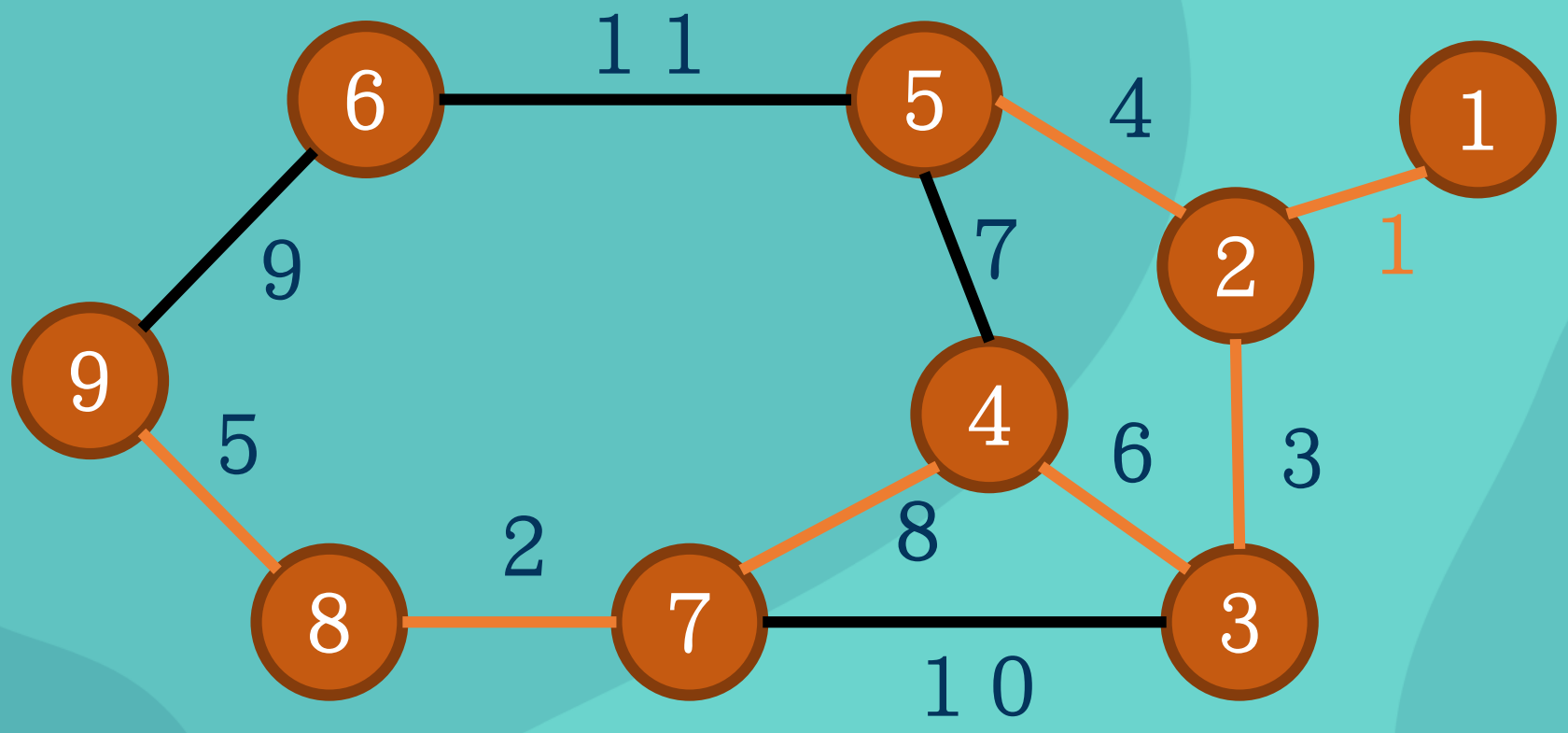

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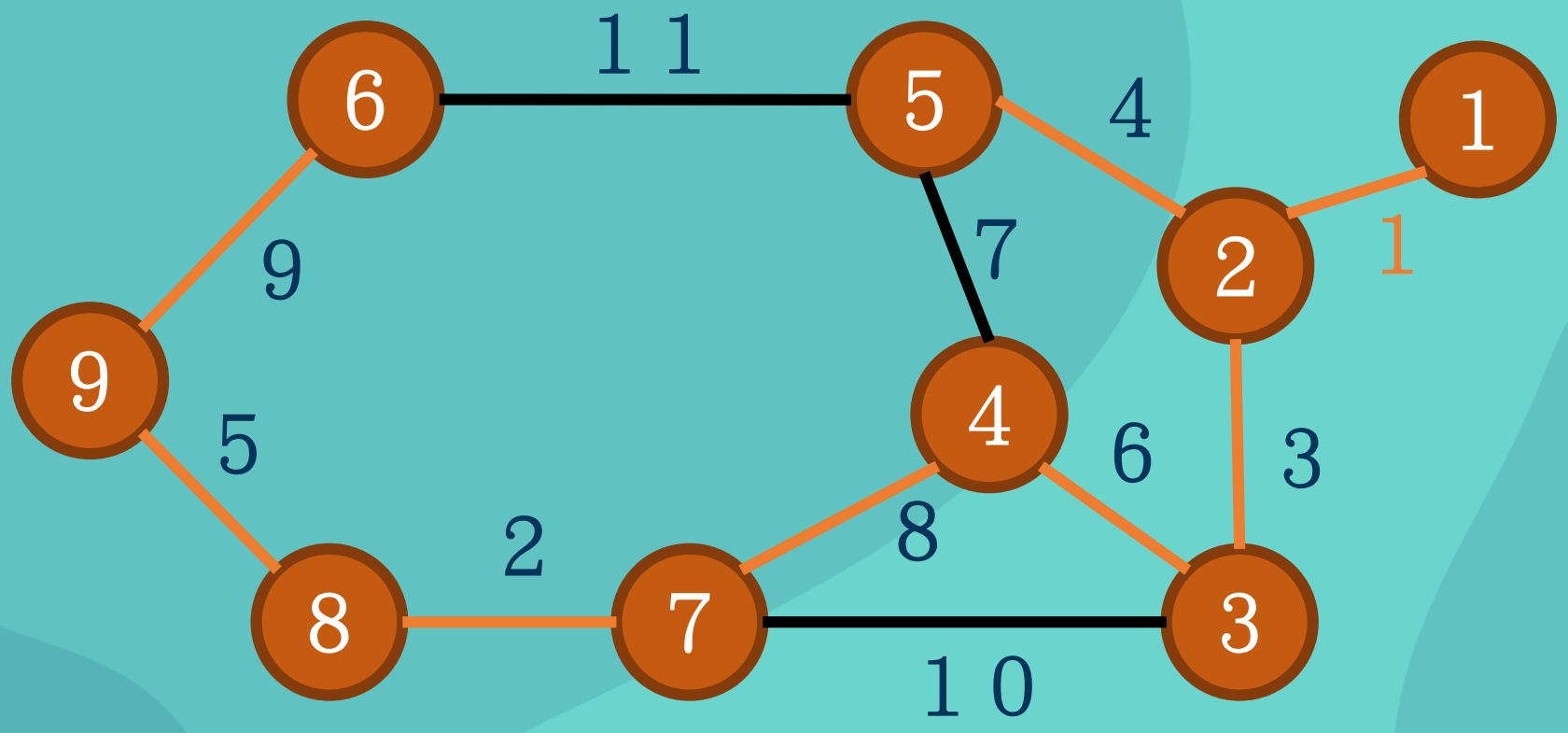

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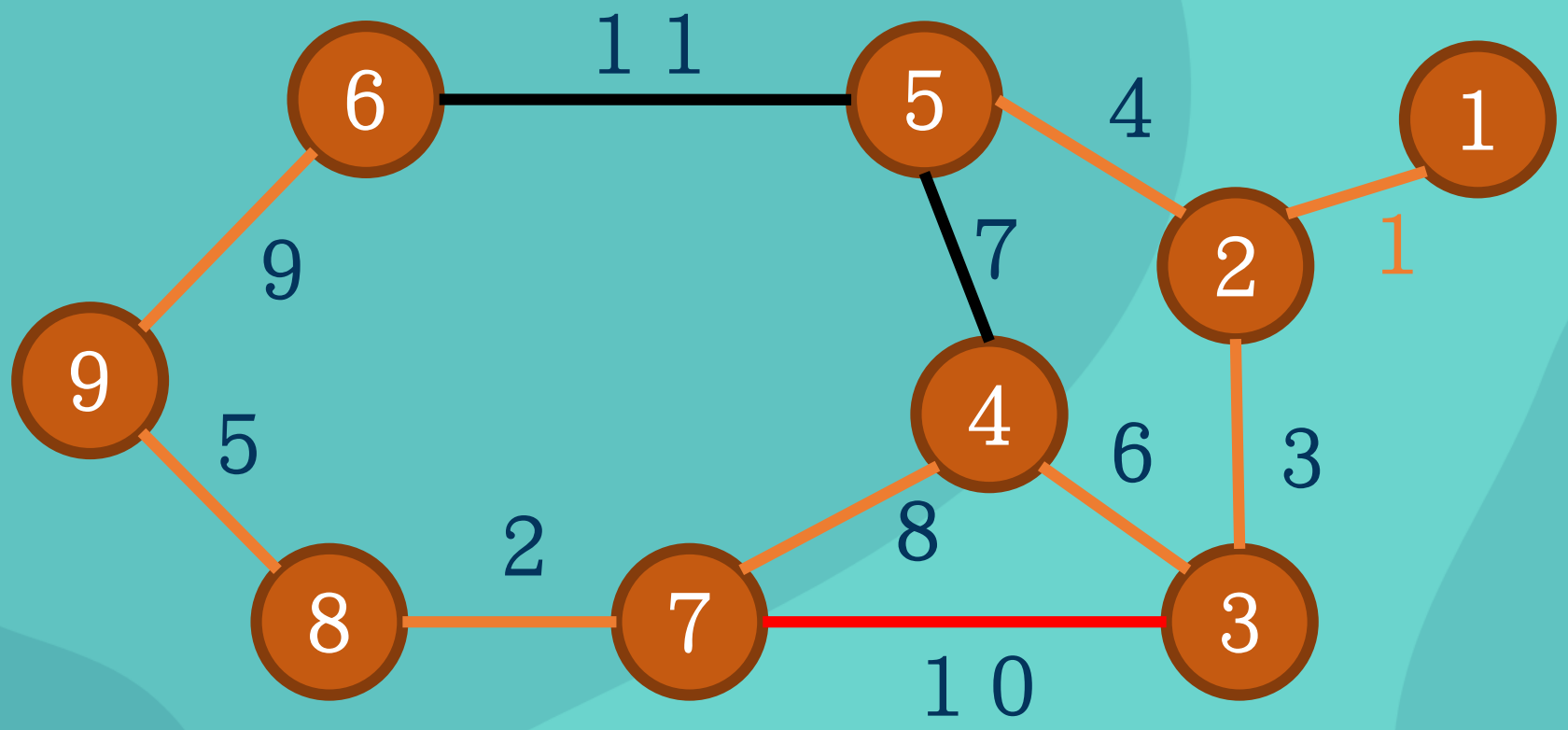

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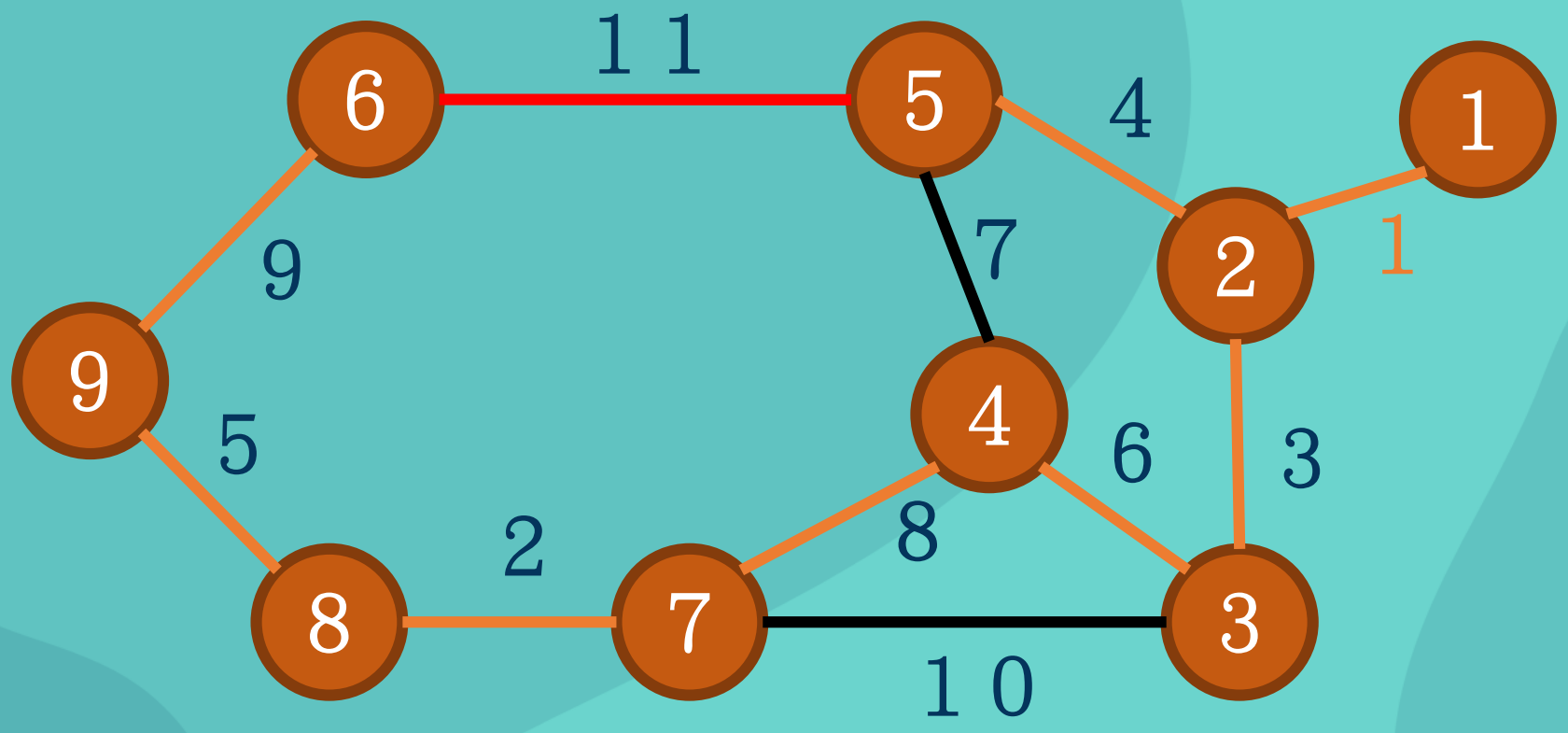
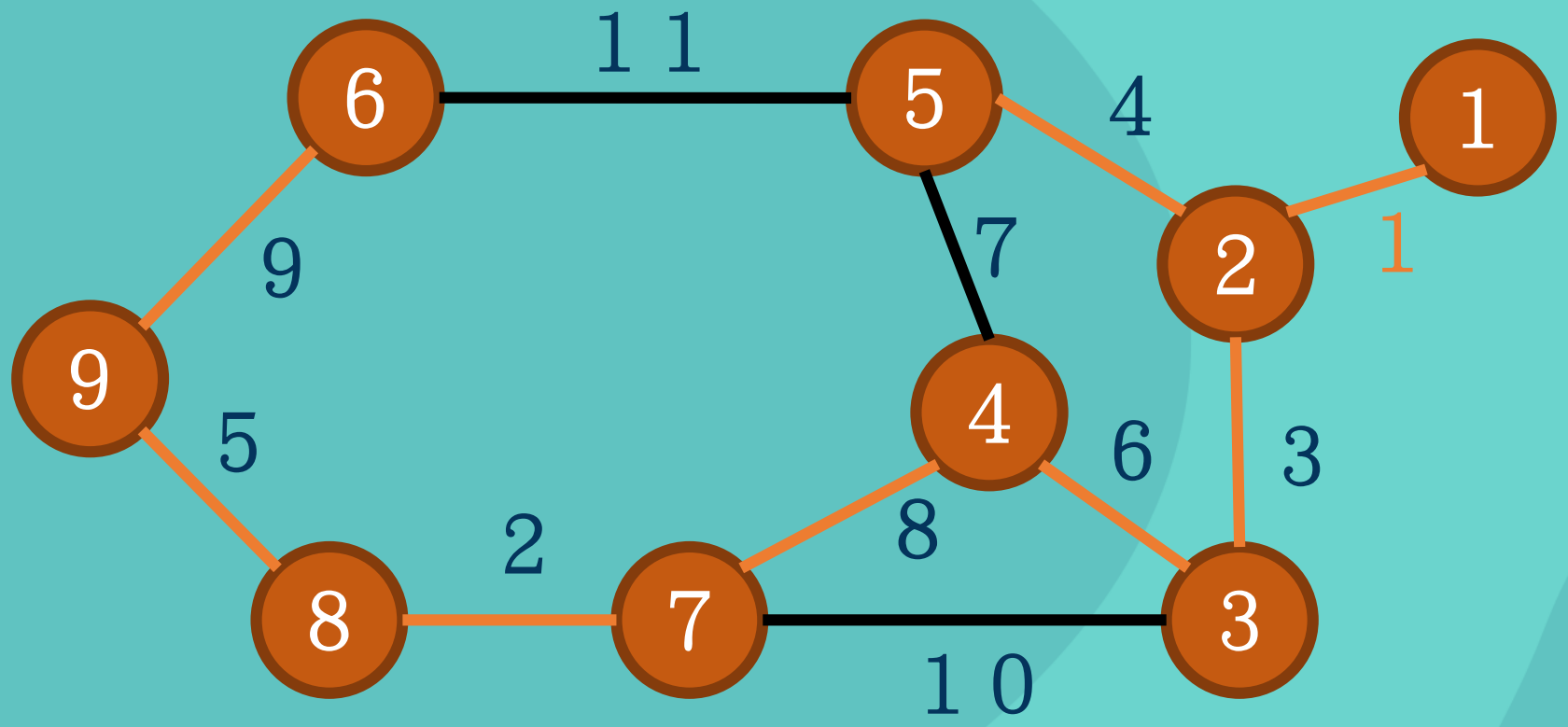
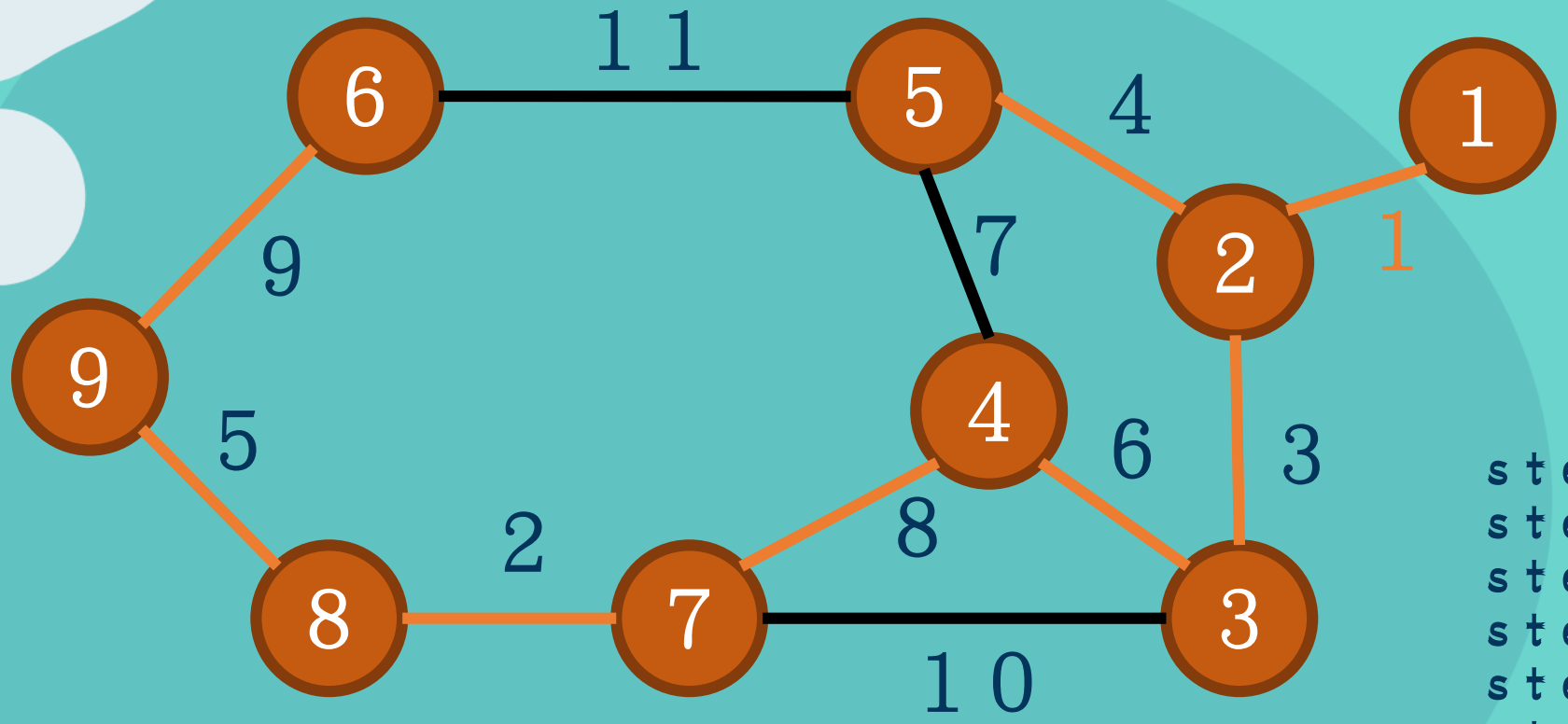





$1+2+3+4+5+6+8+9 = \underline{38\#}$




**總整理**

step1：選擇（1-2）
step2：選擇（7-8）
step3：選擇（2-3）
step4：選擇（2-5）
step5：選擇（8-9）
step6：選擇（3-4）
step7：捨棄（4-5）（會產生cycle）
step8：選擇（4-7）
step9：選擇（6-9）
step10：捨棄（3-7）（會產生cycle）
step11：捨棄（5-6）（會產生cycle）

```
while input(n):

        for 1 to n:
            input(x, y)

        input(m)
        for 1 to m:
            input(a, b)
            smap[a][b] = smap[b][a] = 1;

        r = 0
        for 1 to n
            for 1 to i
                edge[r].point1, point2 = i, j;
                edge[r].weight = (!smap[i][j]) * dist(i, j);
                ++r;

        kruskal(n, r)
```

```
kruskal(int n, int r){
    set_initial(1, n)
    sort(edge)
    sum = 0;
    for 0 to r:
        int A = set_find( edge[i].point1 );
        int B = set_find( edge[i].point2 );

        if(A != B)
            set_union_(A, B)
            sum += edge[i].weight

    return sum;
```